Data Structures & Algorithms (CSC-214) **Semester: Spring 2025**
Assignment 3 **Due date: 26th May 2025, 10 AM**

Submission:
-You are required to use Visual Studio for the assignment. Combine all your work (solution folder) in one .zip file after performing "Clean Solution". Name the .zip file as ROLLNUMBER-A3.zip (e.g. F2024-2345-A3.zip).
-Submit zip file on Google classroom within given deadline.
-Failure to submit according to above format would result in deduction of 10% marks.
Deadline: Deadline to submit assignment is **26th May, 2025, 10 AM**. No submission will be considered for grading outside Google classroom or after this deadline. Correct and timely submission of assignment is responsibility of every student; hence no relaxation will be given to anyone.
Plagiarism: 0 marks will be given if any part of the assignment is found plagiarized. A code is considered plagiarized if more than 20% code is not your own work.

Mode of Testing: Not assigned yet.

Scenario: You are provided with a root directory named charTree. Inside this directory is a binary tree-like structure of folders and subfolders. Each folder can contain up to two subfolders, named left and right, corresponding to the left and right children of a binary tree node.
At each leaf node (i.e., a folder that has no left or right subfolders), there is exactly one .txt file containing alphanumerical characters.

Your task is to:

1. Recursively traverse this directory structure and simulate the following traversals, with alphabets contained within the files (disregard the numbers contained in the .txt file):
   - Depth first traversal: Preorder, Inorder, Postorder Traversal
   - Breadth first traversal
2. Determine the height of the tree
3. Determine the number of text files in the charTree directory
4. Determine whether the tree is height balanced
5. Print only the alphabetical characters in each traversal order as strings.
6. Read and store the alphabetical characters in each traversal order in a file named as "Output.txt".

Considerations:

1. Do not use any external libraries for data structures or algorithms, such as for vector, algorithm, queues. You may use string or cctype for basic string manipulation and fstream

for file handling and any other library for opening the charTree directory. For tree traversal logic, use recursion.

Sample Directory Structure

```
charTree/
├── left/
│   ├── left/
│   │   └── char.txt   (contains 'B')
│   └── right/
│       └── char.txt   (contains 'C')
└── right/
    └── char.txt       (contains 'A')
```

Sample Output:

```
Preorder: B C A
Inorder: B C A
Postorder: C B A
Breadth first:B C A
```

Submission:

- Code file named as described above.
- Screenshot of the sample outputs of the 4 traversals